METHODOLOGY
Advanced Excel Public Procurement & Supplier Risk Dashboard

Project Overview
This project analyses public supplier payments made by Bristol City Council using Advanced Excel. The objective is to monitor supplier spending, identify high-value transactions, analyse payment types, detect potential risk indicators and build an interactive dashboard for procurement control.

Business Problem

Public institutions need to monitor supplier payments to improve transparency, financial control and procurement decision-making. This project addresses the following business question:
How can supplier payment data be transformed into an interactive Excel dashboard to monitor spending patterns, supplier concentration, payment behaviour and potential procurement risks?

Data Source
Source: Bristol City Council supplier payments over £500
Data Type : Public procurement payment records
Period: 2024
Format: CSV files imported and combined in Excel
Main Entity: Bristol City Council
Unit of Analysis: Supplier payment transaction
The dataset consists of public supplier payment records published for transparency purposes. The data was imported from CSV files and consolidated into a single analytical table using Power Query.

Tools Used
Microsoft Excel: Main workbook, dashboard, tables and reporting
Power Query: Data import, cleaning, transformation and consolidation
Power Pivot: Data model, relationships and DAX measures
Pivot Tables: Aggregated analysis by supplier, month, category and payment type
Pivot Charts: Visual representation of spending and risk indicators
Slicers / Timeline: Interactive filtering

Data Preparation Steps
1 - Imported multiple CSV files using Power Query.
2 - Standardised column names to avoid inconsistencies between files.
3 - Cleaned the Amount field and converted it into a numeric format.
4 - Preserved negative values as possible refunds, credits or accounting adjustments.
5 - Created Amount_Clean to represent the signed financial value.
6 - Created Absolute_Amount to analyse the magnitude of transactions.
7 - Created Payment_Type to classify payments and refunds/adjustments.
8 - Created Payment_Size to classify transactions by value range.
9 - Created Expense_Category using the available description fields.
10 - Built dimension tables for date, supplier, expense category, payment size and payment type.
11 - Created relationships in the data model.
12 - Built DAX measures and Pivot Tables for dashboard analysis.

Data Model
The project uses a star-schema style data model, with one central fact table and several dimension tables.
Fact_Payments_Clean: Contains all payment transactions and numeric measures.
Dim_Date: Supports analysis by date, month, quarter and year.
Dim_Supplier: Contains the list of unique suppliers.
Dim_Expense_Category: Contains expense categories derived from description fields.
Dim_Payment_Size: Classifies transactions by payment size.
Dim_Payment_Type: Classifies payments as normal payments or refunds/credits/adjustments.

Key Measures
Total Spend: Calculates net spending using signed payment values.
Gross Spend: Calculates total financial movement using absolute values.
Total Payments: Counts the number of payment records.
Average Payment: Calculates the average payment value.
Highest Payment: Identifies the largest transaction.
Supplier Count: Counts distinct suppliers.
Negative Transactions: Counts refund/credit/adjustment records.
Very High Payments: Counts payments classified as very high.
Top 10 Supplier Spend: Calculates spending concentrated in the top 10 suppliers.
Top 10 Supplier Share: Measures supplier concentration among the top 10 suppliers.

Limitations
The analysis is based on the structure and quality of the publicly available payment records. The dataset does not include detailed contract information, procurement evaluation criteria or supplier performance indicators beyond payment values. Therefore, high-value payments and negative transactions should be interpreted as monitoring indicators, not as evidence of irregularity.

Conclusion
This project demonstrates how Advanced Excel can be used to transform public procurement payment data into an interactive management dashboard. By combining Power Query, Power Pivot, DAX measures, Pivot Tables and dashboard design, the analysis provides a clear view of supplier spending patterns, monthly trends, payment types, high-value transactions and potential supplier risk indicators.